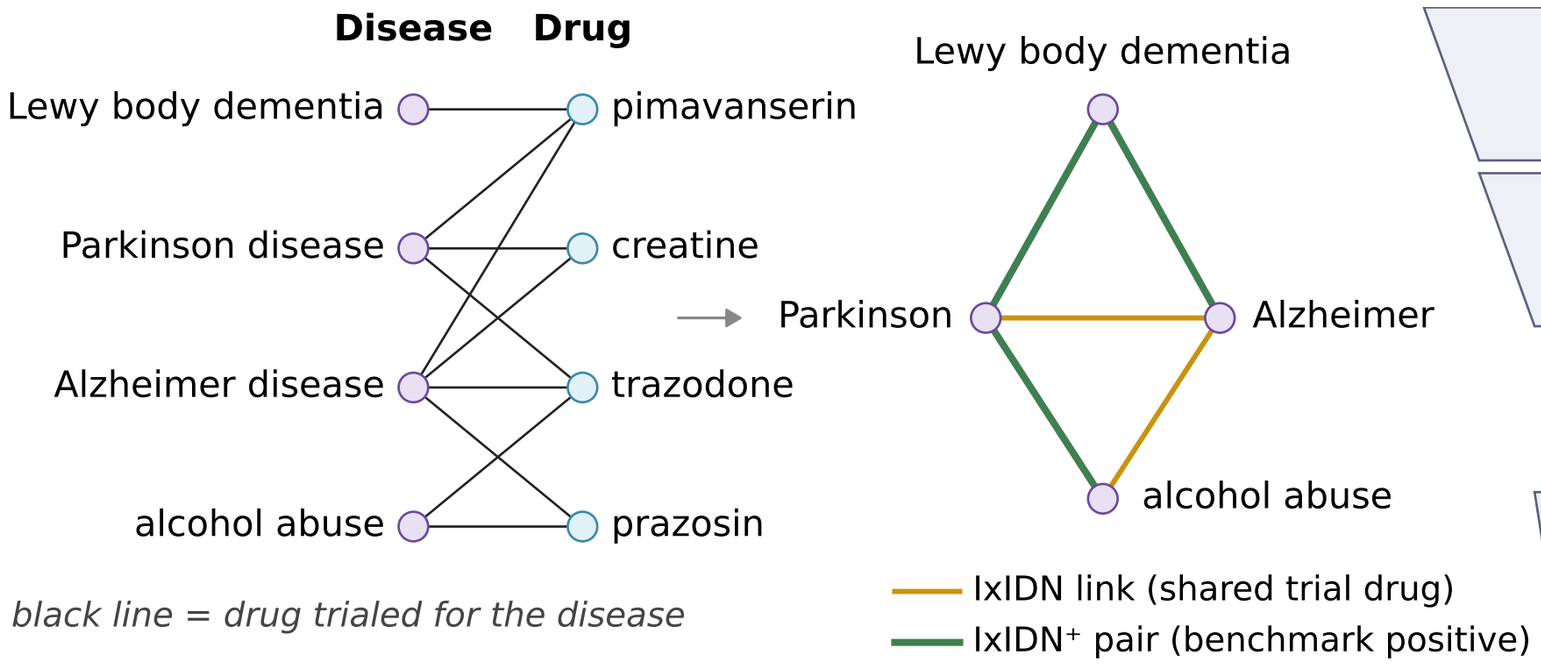
a IxIDN: trial graph projected onto diseases



b From IxIDN to IxIDN-Matched

IxIDN

1,379 diseases, 43,112 IxIDN⁺ links

Evaluation universe

985 diseases

27,415 IxIDN⁺ and 155,375 IxIDN⁻ candidate pairs

Match on 7 potential confounders

1 degree decile, lower-degree disease
2 degree decile, higher-degree disease
3 shared organ system
4 shared Orphanet clinical area
5 name similarity (quartile)
6 MONDO similarity (quartile)
7 gene-coverage category

4,302 IxIDN⁺ pairs with a matching IxIDN⁻

Language-model QC

weak shared-drug evidence dropped
2,182 IxIDN⁺ pairs kept

IxIDN⁺

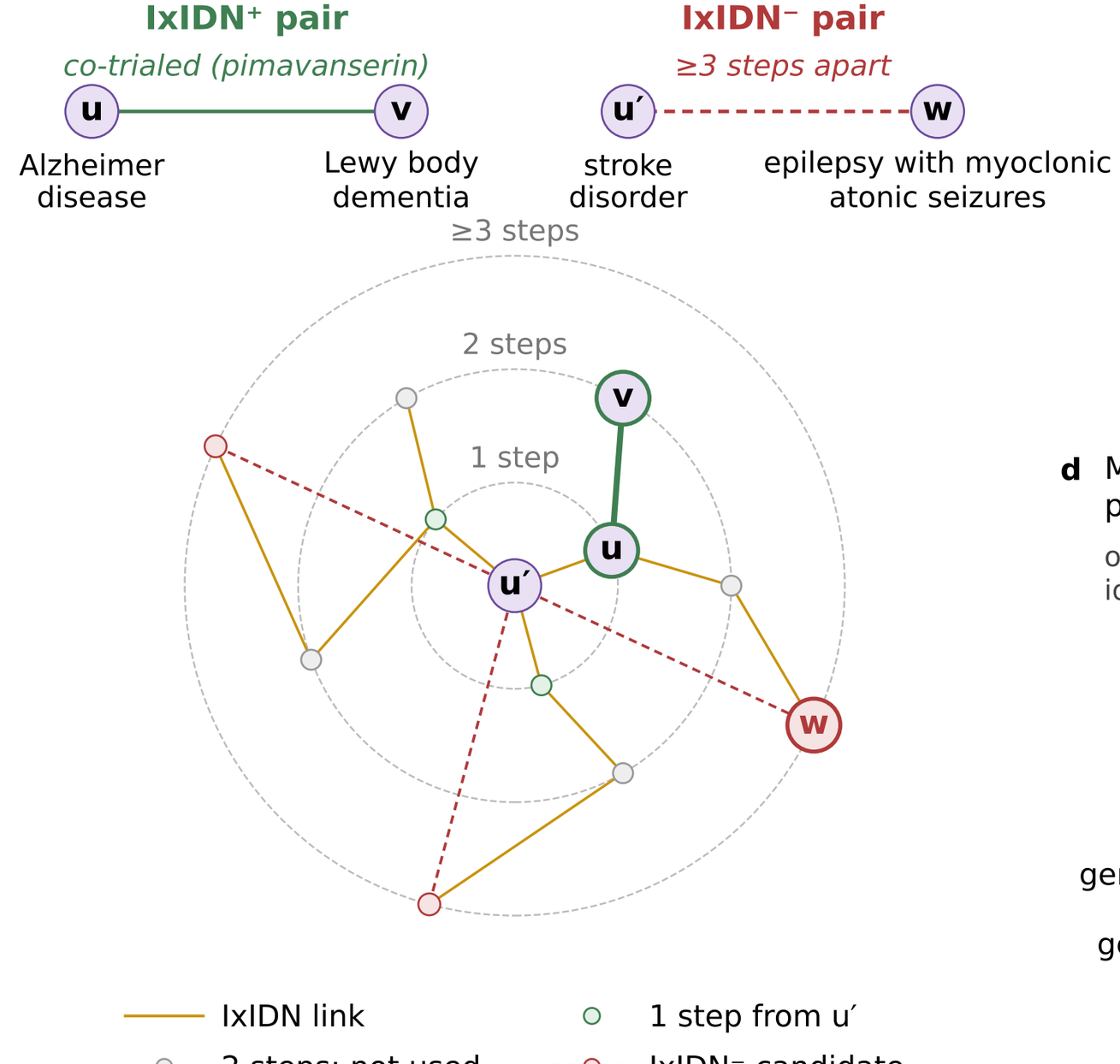
2,182 pairs

IxIDN⁻

2,182 pairs

IxIDN-Matched: 4,364 pairs over 938 diseases

c How an IxIDN⁻ pair is chosen



d Matched confounders alone do not separate positives from negatives

organ system, Orphanet area and gene coverage: identical by construction

